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ORIGINAL RESEARCH

Web Case Points for Early Web Size Prediction

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Abstract
The advent of Internet of Things (IoT) has revolutionized the way information and services are delivered via the web. The widespread popularity of web applications has seen a meteoric rise, with organizations of all sizes, both public and private, opting to make their services and operations available online. However, the development of these web-based applications has become increasingly complex, posing a challenge for organizations to manage this growing complexity within budget and timeline constraints. To address this challenge, there is a pressing need for accurate and efficient web effort estimation methods. The accuracy of such estimation has a direct and profound impact on the survival and success of development companies. This research paper proposes a novel approach to web size estimation, the Web Use Case Points (WCP) method, which takes into consideration functional and non-functional size to predict the size of a web application. The study also compares the WCP method with the widely used Function Point (FP) and Web Object (WO) methods. The results show that the size obtained through the WCP method is smaller than that obtained through the WO method, and greater than that obtained through the FP method in almost all ten web projects analyzed in this study. In conclusion, the WCP method provides a new and promising approach to web effort estimation, offering greater accuracy and efficiency for organizations looking to navigate the complex landscape of web application development.

Keywords Web effort estimation · Web size metrics · Technical and environmental web factors · Web case points

Introduction

The dream conceived by Tim-Berners Lee in 1989 to establish a sophisticated online mechanism for sharing research related data and information among the research community. This was not limited to a group of users but was aimed to connect preachers across bounders and nations. The usage, popularity and demand of this way of information and data communication were widely accepted by different groups with different disciplines. This whole infrastructure

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of sharing and accessing data and information remotely through client/server network architecture is called as World Wide Web or simply as Web. There happened a lot to reform and reframe the way Web used to deliver its services. Data or information made available on web is actually a hyperlinked text build using HTML and now with more sophisticated stuffs like XML, XHTML, Scripts, CMF, and CMS etc. The tradition of delivering content through websites has changed in different perspectives of technology, presentation and accessibility. Web applications are now most used; most demanded and widely discussed interfaces to access and exploit the power of web. Web applications are dynamic, responsive web sites were real-time data interpretation and customization of available constructs is made possible. User stationed at any location connected with communication network can access web resources through graphical user interface contained within web browser.

The popularity of using these web applications has increased explosively and consequently the tradition of using conventional software applications became challenging. The restriction of being within the bounders to access traditional software and now the web applications which is accessible from anywhere made organizations to think beyond rituals. Most of the organizations have switched their operations and services on web by developing their customized web applications. Web application development is different from traditional software development in many aspects [1–5].

The growing demand and much anticipated endorsement of Web Technology is a tire for content delivery in almost all public and private organizations. Increasing popularity and dependence on web based applications have resulted web applications with diverse functionality. This growing heterogeneity, diversity and multitudinous functionality have proportionally increased the complexity of web application development. Web application complexity became a challenge for web project management. Effort estimation is most discussed and very critical challenge for web project management in particular and software project management in general [6]. Web effort estimation is a systematic approach to elicit the efforts required for web application development that particularly corresponds to human efforts. Web effort estimates is very import and most influencing parameter in approximating the cost for web application development. The approximate cost estimation for any web application takes three major constituents into account: Efforts, contingency cost and profit [7]. It is prerequisite for web project management to have an efficient, most reliable and accurate effort estimation process. Accuracy in efforts has direct influence on web Development Company. The more accuracy in estimation results the more beneficial outcomes for web Development Company.

Good effort estimation processes helps project management to understand and draw efficient budgetary schedules for web application development [8]. Efficient results help Development Company to motivate more clients to get their projects developed. This raises Development Company high in competition and consequently outstanding figures in revenue and profit. Less efficient and inaccuracy in effort estimation results have negative repercussions on web development company [9]. This is now inevitable and indispensable for web project management to design and development effect and effective effort estimation process so, accurate effort estimation results can be made available before any actual development is initiated.

This study is an attempt to propose web effort estimation approach. In this study new web size measurement metrics: web use point (WCP) is introduced. WCP is adjusted size of web application that represents both functional and dimensional size measures required for web application development. The remainder of this paper is organized as follows. Section "Web Effort Estimation" gives brief discussion of web effort estimation models, section "Background" discusses background of web size metrics, section "Web Case Points" gives overview of proposed size metrics, WCP, in section "Results and Discussions" comparative analysis of WCP, WO and FP is performed and finally section "Conclusion" the conclusion.

Web Effort Estimation

As mentioned above, web effort estimation is a systematic procedure used to approximate or forecast the extent of efforts required to carry out a successful web application development. Effort estimation models can be broadly categorized as algorithmic models and non-algorithmic models. Algorithmic models use certain empirical or mathematical relationship model between certain dependent and independent variables whereas non-algorithmic models use past project data and expert based judgment to decide efforts for web application development [10–13]. To offer best cost for web application development, web development company needs to have more appropriate method or technique to obtain realistic cost for web application development [9]. The cost of development have direct relationship with web application size which intern depends on various requirements to be delivered by web application to be developed [14]. The perfectness in requirement elicitation and their analysis possess visible reflections on effort estimation. The standard and systematic approach that is considered during effort estimation of any web application development is shown in Fig. 1.

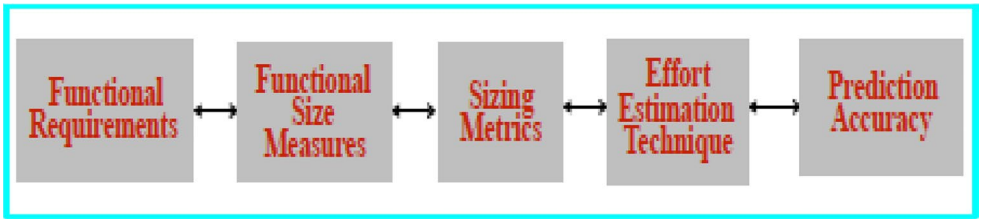

Fig. 1 Abstract view of web effort estimation

As obvious from above figure, web effort estimation is not a solo process but is actually an integrated approach of different specialized sub process. Each of these steps forward more relevant information in order to acquire best and accurate effort estimates as outcome.

Effort estimation for web application development is usually carried out by exploiting traditional effort estimation models. There is no doubt that their implementation has produced good estimation results in certain web applications but as a generalized approach they failed to draw better estimation inferences. The inaccuracy in effort estimates that was observed is directly tribute to implementation of ad-hoc or traditional approaches for web effort estimation [15]. There has been many effort estimation methodologies developed from time to time by different researchers. These methodologies were aimed to fulfill the challenges faced by project management during effort estimation. Few popular functional size measurement metrics used to obtain web development size and consequently the efforts were FP [16], WO [9], RWO [17], CFSU [18], UCP [19], Internet Points [20], Web Point Metrics [21], Data Web Points [22], etc. Web objects proposed by Donald, J. Reifer are the first web specific functional size metrics [9]. This functional size measurements quantification was done on the basis of functional user requirements specified for particular web application development. These proposed approaches didn't succeed much. There still continues the implementation of fractional and ad-hoc approaches for web effort estimation.

Background

A functional and dimensional measure of web application development constitutes the size of web application. The identification of these measures is very critical for effective size prediction and consequently the accurate web effort estimates. Web effort estimation was carried out by using different available traditional software effort estimation models. With the increasing diversity in architecture, functionality, content, navigational structure, behavior and implementation, web application complexity increased many folds. It was pretty challenging for web project management to render the web development perfectively by extending to traditional practices and ad-hoc methods [15]. To develop new approaches to ensure effectiveness in web application development in general and web effort estimation in particular, different models were proposed. Most of these approaches were practices to obtain functional size measures of web application and very few prediction models specific to web effort estimation were developed. The literature reviewed in [10] generates direct and clear insights that there is not even a single model available in literature that performs web effort estimation accurately and effectively [23].

Donald, J. Reifer [9] proposed web objects web size measures after adding four web specific components to traditional function points approach and WebMo were developed as effort prediction model and no empirical analysis were discussed for WebMo. Emilia Mendes et al., performed web size measures using attributes: length, functional and complexity measures [24]. Leonardo M and Roberta [25] developed metrics model based on W2000, size measured from functional, navigational, structural and multimedia components. Sergio, F. et al., proposed data web points to measure functional size of web [22]. Melanic Ruhe et al., compared FP and WO the results revealed that WO with OLS performed better than FP and also revealed size expressed in WO were 55% more than FP [26]. Paul Umber [27] expressed web size as CFSU and proposed COSMIC-FFP performed better effort prediction than expert-based judgment.

Costagliola, G. et al., [28, 29] in his study revealed that COSMIC-FFP produced better effort prediction results. Costagliola in his other study divided size measures into functional measures and length measures, WO were used as web size metrics the results revealed that the proposed approach played better estimates than analogy based approaches [30]. Ferrucci et al., investigated COSMIC and WO, the results showed COSMIC as better indicator [31]. Giulio, B. Giulio et al., studied the effectiveness of FP and WO, as web size measures and showed that WO performed better in comparison to FP [32]. Filament Ferrucci [33] studied Web-CORBA and COSMIC size measures and the results revealed were better than [34].

Silvia Abraham used OO-HFP and was more accurate than FPA [35]. Sergio, D. Martino performed replicated study and showed WO better than FP when OLSR, Web-CORBA were used as predictive approaches [36]. R. Folgieriet et al., in his study proposed RWO and results showed that RWO performed better in comparison to FP and WO

[9]. Erika Corona et al., opposed web CMF model for web effort estimation and results showed better effort prediction than RWO and FP [37]. Lucio De Marco performed web size approximation by COSMIC functional process (cfunP) and average functional process(AFP), the results obtained were significant [38]. Similarly Denis Cake et al., proposed hybrid approach by clubbing COSMIC-FFP and UWE, the predictability results obtained were satisfying Contest Criteria [39]. The detailed review on these studies can be obtained from [10].

Use Case Point (UCP) analysis has been widely employed as an early-stage effort estimation technique for software development, offering a structured approach grounded in functional requirements. Numerous studies have extended and enhanced the traditional UCP model to address its limitations and improve its prediction accuracy.Ramesh and Varalakshmi (2023) proposed an ensemble learning-based prediction model for UCP-based effort estimation. Their study demonstrated that machine learning ensembles such as Random Forest and AdaBoost significantly improve estimation accuracy compared to conventional UCP approaches. The findings emphasize the benefits of integrating intelligent learning methods into traditional estimation techniques [40]. In a systematic review, Rani and Arora (2021) analyzed the existing literature on UCP-based effort estimation. They observed a growing trend toward hybrid and model-enhanced estimation approaches and highlighted key challenges such as adjusting for productivity rates and calibration factors. Their work provides a strong foundation for understanding the evolution of UCP applications over the past decade [41]. To address the inherent uncertainty in early-stage estimation, Vetriselvan and Suresh (2024) introduced an Adaptive Neuro-Fuzzy Inference System (ANFIS) integrated with Enhanced Use Case Points (EUCP). Their hybrid model accounts for imprecise and fuzzy project attributes, resulting in better adaptability and prediction accuracy, particularly during the requirements phase [42]. Another refinement in UCP modeling was proposed by Febriani and Hidayat (2023), who incorporated activity-based function point complexity into use case diagrams. Their approach offers a more granular analysis of software activities, thereby refining the estimation of development effort and aligning it more closely with actual project tasks [43]. In the context of software reuse, Haron et al. (2019) investigated effort estimation models that utilize reusable use cases. Their case study confirmed that use case reuse can lead to improved efficiency and estimation accuracy, especially for projects with similar functional scopes. This approach supports the development of reusable software components as a strategic asset in effort estimation [44].

The insights from above discussion about different size metrics and their behaviour in web effort estimation have been quite elusive to find more accurate choice. Nevertheless it was possible to observe the popularity and usability of web objects and function points to express the functional size of web application development. Selection of parameters and their mapping as web size metrics have influence on the accuracy of efforts estimated [45, 46]. The need of the hour is to propose new and affect approach for pursing effective web effort estimation. To over these challenges this study proposes new web size metrics, web case pints to predict web application size.

Table 1 WCF with their fixed weighting factor

Factor	Description	Weight(w)	Count(C)
W1	No. of web Pages	0.5	
W2	No. interactive web pages	2	
W3	No. of multimedia Files	1	
W4	No. of application points	1	
W5	Multilingual support	1	

Web Case Points

Web case points (WCP), a web size metrics proposed in this study are based on functional, dimensional and nontechnical parameters required for web application development. WCPs were obtained from requirements elicited and elaborated through objective oriented model-driven approach called as use case modelling. In use case modelling, use case diagrams are used to express requirements into their corresponding functional and dimensional functional measures. Actors, use-cases, use case scenarios and different type's relations and associations are main constructs in objective oriented approach to model working framework of web application development. The overall size of the web application development is carried out in multiple steps:

Step I: Obtaining Web Complexity Factors corresponding to functional size

The actual size expressed as WCPs is calculated from web complexity factors (WCF) proposed in [47]. WCF has five parameters to represent functional and dimensional aspect of web development: no. of web pages, no. of interactive web pages, no. of application points, no. of multimedia files and nature of multilingualism. The identification, behaviour and implementation of operands and operators are carried out through use case modeling. Each of these parameters has associated a fixed weighting factor. This weighting factor relates to their complexity. The overall density of these parameters produce unadjusted size of web application calculated as unadjusted web case points by using equation no. 1 and parameters are shown in Table 1 below.

$$UWCP \sum_{i=1}^{5} (W_i C_i) \quad (1)$$

In order to normalize the unadjusted weight the factors that influence WCP were identified as technical and environmental web complexity factors as were used by Karnar in software effort estimation [19]. Both technical and environmental factors in [19] were revisited to find their relevance with web application development. The relevance was obtained from respondents who were researchers, academicians, web development practitioners, consultants and freelancers through questionnaires and interviews. The trends of the reviewed responses were in favor of inclusion of ECF and TCF of [19, 48] after certain changes.

Step II: Identification and Inclusion of Technical Complexity Factors (TCF_{web})

The Technical Complexity Factors were extended from the UCP model to reflect web-specific technical challenges more accurately. Fifteen (15) factors were included in the proposed TCF_{web}, based on their expert-rated relevance. These include original factors like "Complex Internal Processing" and "Code Reusability," as well as web-specific refinements such as "Web Development Architecture," "Scalable and Reliable Systems," and "Database Integration" as shown in Table 2 below:.

The TCF_{web} is calculated using equation :

$$TCF_{\text{web}} = 0.6 + (0.01 \times \sum_{i=1}^{15} W_i S_i) \quad (2)$$

Where: W_I is the fixed weight of the technical factor and $S_i$is the significance score (0–5) based on expert input.

This factor adjusts the UWCP to reflect the technical rigor and infrastructure needs associated with the web application.

Step III: Identification and Inclusion of Environmental Complexity Factors (ECF_{web})

The Environmental Complexity Factors (*ECF_{web}*) encapsulate the development team's characteristics, project methodology, and work environment. A total of 10 factors were finalized after evaluating expert responses, including "Familiarity with Development," "Use of Development Tools," and the newly added factor "Testability." Table 3 lists these factors along with their corresponding weights.

The *ECF_{web}* is computed using:

$$ECF_{\text{web}} = 1.4 + (-0.03 \times \sum_{i=1}^{10} W_i S_i) \quad (3)$$

Table 2 Technical complexity factors TCFweb (T1-T15) for web application development

Factor	Description	Weight(w)	Significance(0, 5)
T1	WebApp development Architecture	2	
T2	Persistence and throughput	1	
T3	End User Efficiency	1	
T4	Complex Internal Processing Required	1	
T5	code reusability	1	
T6	Installation Ease	0.5	
T7	Usability	0.5	
T8	Cross-Platform Support	2	
T9	Easy To Change	1	
T10	Highly Concurrent	1	
T11	Custom Security	1	
T12	Dependence On Third-Party Code	1	
T13	User Training	1	
T14	Scalable and Reliable	**2**	
T15	Database integration.	2	

Table 3 Environmental complexity factors (E1-E10) for web application development

Factor	Description	Weight(w)	Significance(0, 5)
E1	Familiarity With The web application development.	1.5	
E2	Developers technical capability	0.5	
E3	OO Programming Experience	1	
E4	Lead Analyst Capability	0.5	
E5	Motivation	1	
E6	Stable Requirements	2	
E7	Usage of developmental tools(CMS/CMF support)	-1	
E8	Difficult Programming Language	-1	
E9	Project Methodology	1	
E10	Testability	0.5	

Where W_i is the fixed weight for each factor, and $S_i$is the expert-assigned significance score.This factor adjusts for productivity variations due to team capability, technology familiarity, and development practices.

Step IV: Calculating the overall Size expressed in Web Case Points (WCP's)

After obtaining UWCP, TCFweb and ECFweb, the next component for web size estimation is to calculate adjusted web application size that is to get total number of web case points (WCP) or adjusted web case points (AWCP). The actual causation of WCPs is expressed by using Eq. 4a or 4b

$$WCP = UWCP \times TCF_{\text{web}} \times ECF_{\text{web}} \quad (4a)$$

$$AWCP = UWCP \times TCF_{\text{web}} \times ECF_{\text{web}} \quad (4b)$$

Table 4 Estimating total number(size) of FP, WO and UWCP calculated from individual project

Project ID	FP	WO	WCP
P1	41	79	48
P2	83	208	132
P3	127	269	174
P4	234	424	320
P5	218	390	291
P6	166	303	202
P7	218	404	286
P8	83	159	88
P9	221	481	288
P10	349	434	344

Table 5 Magnitude of relative error in estimated efforts obtained by using FPA, WebMo and Web-UCP

Project ID	Magnitude of Relative Error(MRE)		
	FPA	WebMo	Web-UCP
P1	0.30282	0.20423	0.10915
P2	0.15634	0.02065	0.02507
P3	0.25056	0.20449	0.03146
P4	0.11430	0.08206	0.01172
P5	0.09275	0.03560	0.01655
P6	0.09076	0.04538	0.06975
P7	0.11030	0.07758	0.08485
P8	0.20209	0.03484	0.05749
P9	0.13049	0.11093	0.09057
P10	0.07785	0.03509	0.02166

Where WCP or AWCP is estimated size of web application and is expressed as total number of web case points after considering the impact of technical and environmental factors on web development, UWCP are unadjusted web case points, TCFweb is technical complexity factor and ECF-web, is environmental complexity factor.

Step V: Integration and Final Effort Estimation

After computing the WCP, the final effort estimation is determined by factoring in the Web Application Complexity Ranking (WAC_{rank}), which classifies the application as Simple, Average, Complex, or Critical with respective multipliers of 5, 10, 15, or 20. The final estimated effort is then:

$$Effort = WCP \times WAC_{rank} \tag{5}$$

$$Effort\,(\text{Person} - \text{Hours}) = \text{WCP} \times \text{WAC}_{\text{rank}} \tag{6}$$

Where *WCP* is total calculated web application size and WAP_{rank} is a fixed web application complexity factor determined by using expert-based judgment approach. *Efforts* are estimated efforts for web application development and are expressed in *person hours.*

Results and Discussions

The proposed web size metric, WCP, was rigorously evaluated in this study by comparing its estimated size with two widely used metrics, FP and WO. Both FP and WO calculate the functional size of a web application. FP, a commonly used sizing metric for software effort estimation, and WO, the first web-specific size metric, were implemented for effort estimation as described in references [11] and [9].

A dataset of 10 industrial web projects was employed to assess the accuracy of FP, WO, and WCP in calculating web application size. The results of the implementation are presented in Table 4.

The results indicate that the size estimates obtained by FP were smaller than both WO and WCP in the majority of the 10 web projects. WO, on the other hand, produced larger size estimates than FP and WCP. The proposed WCP metric was found to provide a balanced size estimate that falls between the estimates obtained by FP and WO.

This study highlights the limitations of FP in web effort estimation and underscores the need for a more comprehensive metric that takes into account all aspects and parameters relevant to web application development. While WO was specifically designed for web effort estimation, the results show that WCP provides more balanced size estimates in most of the web projects studied.

In conclusion, this study demonstrates the potential of WCP to provide better size estimates for web application development compared to commonly used FP and WO metrics. The balanced size estimate obtained through WCP can lead to more accurate effort estimates, making it a valuable addition to the field of web effort estimation.

Performance Evaluation Metrics

To assess the performance of Web-UCP, WebMo, and FPA for web effort estimation, this study applied standard evaluation metrics—including MRE, MMRE, MdMRE, MAE, MdAE, STD DEV, and PRED(25/20/10)—on a dataset of ten projects. The results from Web-UCP were then compared with those of WebMo and FPA to evaluate its relative accuracy and effectiveness.

Magnitude of Relative Error(MRE)

Magnitude of Relative Error (MRE) for Web-UCP, WebMo, and FPA was calculated using Eq. 7 across ten web projects (P1–P10), as presented in Table 5.

$$MRE = \frac{Effort_{Actual} - Effort_{Estimated}}{Effort_{Actual}} \tag{7}$$

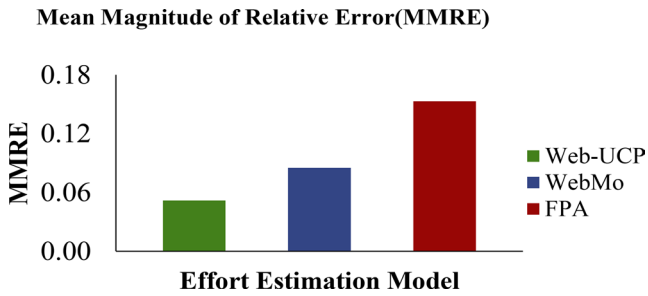

Fig. 2 MMRE in calculated efforts obtained after using Web-UCP, WebMo and FPA

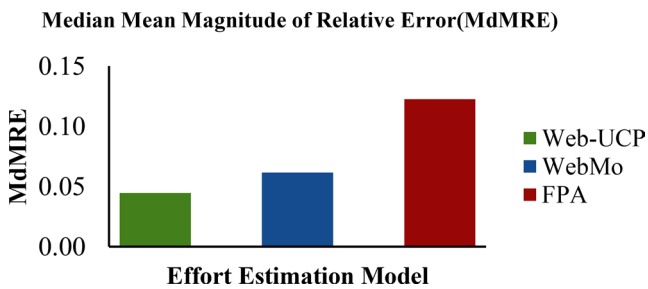

Fig. 3 MdMRE in calculated efforts obtained after using Web-UCP, WebMo and FPA

Table 5 shows that MRE values for Web-UCP are consistently lower than those for WebMo and FPA, highlighting the superior accuracy of Web-UCP in web effort estimation.

Mean Magnitude of Relative Error (MMRE)

Mean Magnitude of Relative Error (MMRE) was calculated from the MRE values for projects P1–P10 using Eq. 8. It quantifies the average relative deviation in estimated efforts, regardless of units (e.g., person-hours or man-months), and is illustrated in Fig. 2. Deviations may be either underestimates or overestimates.

$$MMRE = \frac{1}{n}\sum_{i=1}^{n} MRE \tag{8}$$

Figure 2 shows that MMRE values for Web-UCP are lower than those for WebMo and FPA, indicating that Web-UCP provides more accurate effort estimation. WebMo also outperforms FPA in this regard.

Median Magnitude of Relative Error (MdMRE)

The Median Magnitude of Relative Error (MdMRE) represents the median relative deviation between estimated and actual efforts. Figure 3 shows the MdMRE for projects P1 to P10 using Web-UCP, WebMO, and FPA.

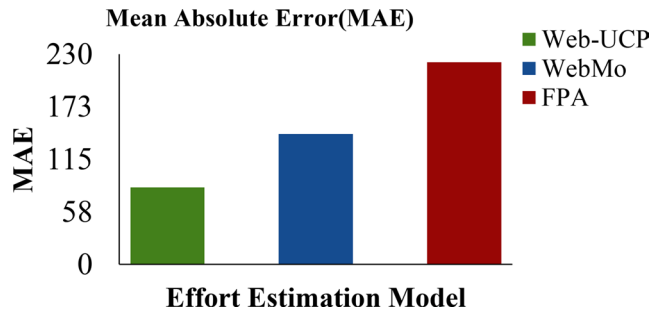

Fig. 4 MAE in calculated efforts obtained after using Web-UCP, WebMo and FPA

The perusal of Fig. 3 shows MdMRE for web projects(P1-P10) obtained using Web-UCP is comparatively lesser than the MdMRE obtained using WebMo and FPA. Similarly MdMRE for projects(P1-P10) obtained using WebMo is less than MdMRE obtained using FPA.

Mean Absolute Error (MAE)

Absolute error measures the difference between estimated and actual efforts. The Mean Absolute Error (MAE) is the average of these absolute errors for projects P1–P10, calculated using Web-UCP, WebMo, and FPA. MAE is computed by Eq. 8 and shown in Fig. 4.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |Effort_{Actual} - Effort_{Estimated}| \tag{9}$$

Figure 4 clearly shows that the MAE for projects (P1–P10) using the Web-UCP model is lower than that of WebMo and FPA. Additionally, WebMo performs better than FPA for web effort estimation. These results highlight the superior performance of the Web-UCP model compared to WebMo and FPA.

Mean, Standard Deviation(STDEV) and Variance of Estimated Efforts

The calculated efforts were further analyzed by comparing the mean, standard deviation, and variance resulting from Web-UCP, WebMo, and FPA for web effort estimation, as shown in Figs. 8, 9 and 10, respectively. The mean actual effort for projects P1–P10 is approximately 1926 man-hours. The mean estimated efforts using Web-UCP, WebMo, and FPA are 1851, 1800, and 1698 man-hours, respectively. The mean estimated by Web-UCP is closer to the actual mean effort than those from WebMo and FPA, as illustrated in Fig. 5.

Figure 5 shows the standard deviation of estimated efforts using Web-UCP, WebMo, and FPA. The standard deviation from Web-UCP (1353.061) closely matches that

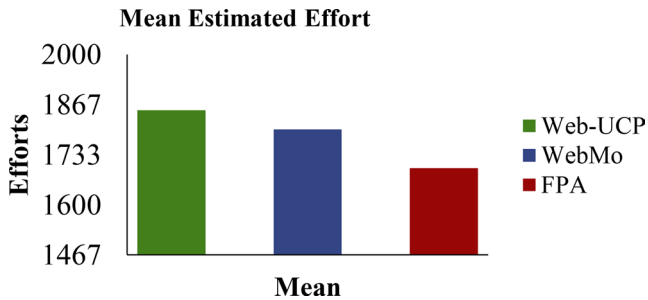

Fig. 5 Mean estimated efforts obtained using Web-UCP, WebMo and FPA

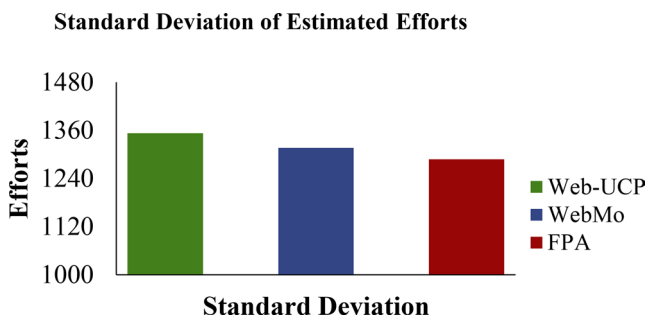

Fig. 6 STDEV in estimated efforts obtained using Web-UCP, WebMo and FPA

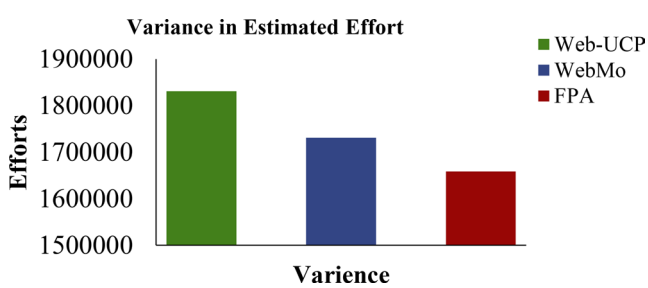

Fig. 7 Variance of calculated efforts after using Web-UCP, WebMo and FPA

of the actual efforts (1407.76). In comparison, the standard deviations for WebMo and FPA are 1315.577 and 1287.952, respectively [49].

Based on the standard deviation results, Fig. 6 compares the variance of estimated efforts using Web-UCP, WebMo, and FPA with the actual efforts variance (1,981,781). It shows that the variance from Web-UCP (1,830,773.7) closely aligns with the actual efforts variance (See Fig. 7).

The overall interpretations obtained from Mean, Standard Deviation and Variance in effort estimated using Web-UCP, WebMo and FPA for projects P1 through P10 shows better performance and close relevance of Web-UCP in comparison with WebMo and FPA for web effort estimation.

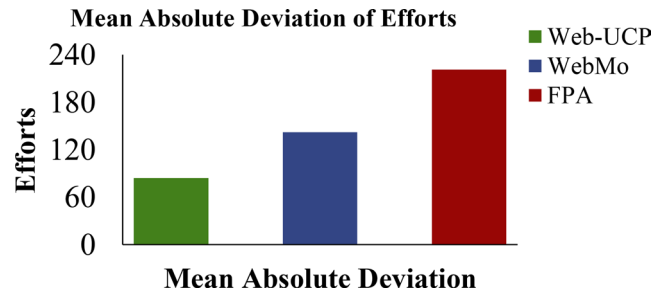

Fig. 8 Mean of deviations in estimated efforts obtained using Web-UCP, WebMo and FPA

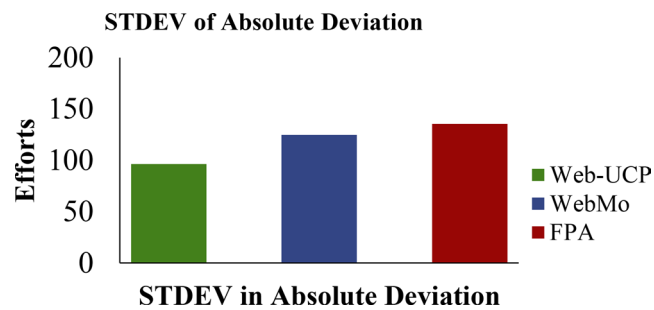

Fig. 9 STDEV of absolute deviations in estimated efforts obtained using Web-UCP, WebMo and FPA

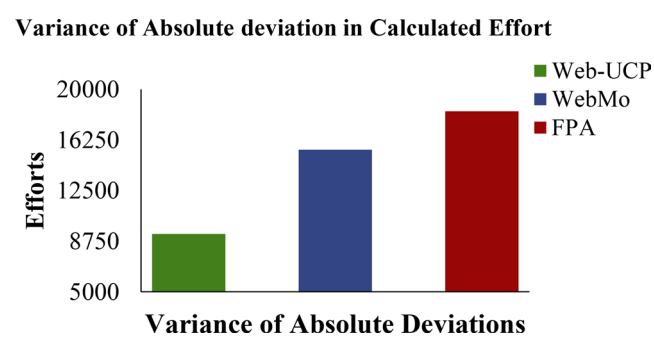

Fig. 10 Variance of deviations in estimated efforts obtained using Web-UCP, WebMo and FPA

Mean, Standard Deviation (STDEV) and Variance of Deviation in Estimated Efforts Estimated Using Web-UCP, WebMo and FPA

In order to further evaluate the effectiveness and accuracy in results obtained using Web-UCP, WebMo and FPA, this study performs Mean, Standard deviation and Variance on the absolute deviation in estimated efforts. Absolute deviation quantifies the gap between actual and estimated efforts, more gap mean more deviation. Mean, Standard deviation and Variance calculated is graphically expressed by Figs. 8, 9 and 10 respectively.

The perusal of the results expressed in Fig. 8 show that mean of deviation using Web-UCP were comparatively less than mean of deviations obtained using WebMo and FPA for web effort estimation.

Similarly, the STDEV of deviation obtained using Web-UCP model were comparatively lesser than STDEV of deviations in efforts estimated using WebMo and FPA and the same is given in Fig. 9.

Further the variance of deviation obtained using Web-UCP, WebMo and FPA for projects P1-P10 reiterates the better performance of Web-UCP in comparison with WebMo and FPA models for web effort estimation and the same is given in Fig. 10.

The results from evaluating Web-UCP, WebMo, and FPA across various criteria clearly demonstrate that the Web-UCP model significantly outperforms WebMo and FPA in estimating web effort for all projects P1 through P10. Additionally, the findings show that WebMo performs better than FPA. Overall, these results strongly support the superior performance of the Web-UCP model for web effort estimation.

Predictability (PRED(n))

To further evaluate the results, effort predictability was evaluated. Predictability, or *Pred*(x), is defined as the number or proportion of projects with an MRE less than or equal to the prediction level 'x'. *Pred*(x) is calculated using Eq. 10.

$$\text{Pred}\ (x) = \frac{T}{N} \tag{10}$$

Where N is total number of projects, T represents total number of projects where calculated MRE of projects P1, P2 upto P10 is less than or equal to "x". *Pred*(x) is the total number of projects where MRE less than or equal to the value substituted for variable "x".

In this study, *Pred*(25), *Pred*(20) and *Pred*(10) is performed on efforts estimated using Web-UCP, WebMo and FPA and the results obtained are given in Figs. 11, 12 and 13 respectively.

Figure 11 presents the *Pred*(25) results for estimated efforts of projects P1 to P10. The data clearly show that *Pred*(25) using Web-UCP is higher compared to WebMo and FPA. Additionally, WebMo's *Pred*(25) outperforms FPA. These results also meet Conte's Predictability Criteria, which requires *Pred*(25) to be at least 75%, meaning that 75% or more of the projects have an MRE of 0.25 or less [24].

Similarly, the results obtained for *pred*(20) are given in Fig. 12 and shows that Web-UCP performed comparatively better than WebMo and FPA in projects P1, P2 upto P10.

The *Pred*(10) results in Fig. 13 further confirm the superior performance of the Web-UCP model compared to WebMo and FPA for web effort estimation. Overall, the predictability values of Web-UCP at *Pred*(25), *Pred*(20), and *Pred*(10) meet Conte's Predictability Criteria and outperform those of WebMo and FPA. Additionally, the predictability results for WebMo are consistently better than those for FPA. These findings clearly demonstrate the better performance of Web-UCP over WebMo and FPA in web effort estimation.

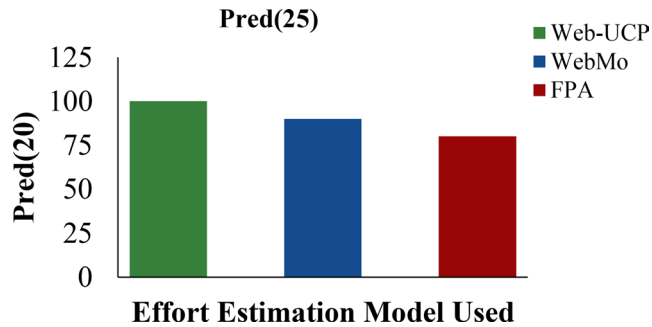

Fig. 11 Pred (25) of calculated efforts using Web-UCP, WebMo and FPA

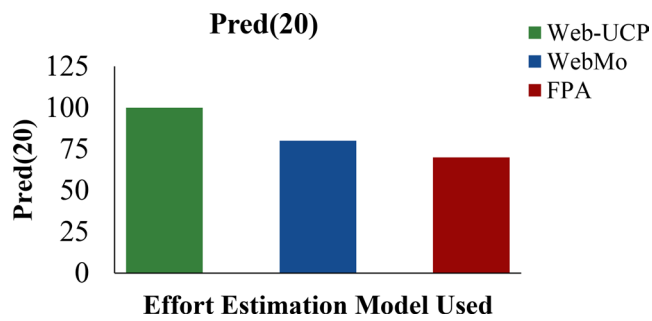

Fig. 12 Pred(20) of calculated efforts using Web-UCP, WebMo and FPA

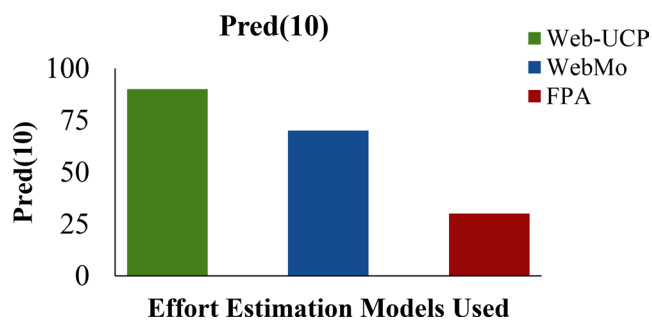

Fig. 13 *Pred*(10) of calculated efforts using Web-UCP, WebMo and FPA

Conclusion

In this study, a new web size metrics was proposed to calculate the size of web applications, the Web Use Case Points (WCP) method. The WCP method is based on three different complexity categories that reflect the functional, dimensional, technical, and environmental parameters that impact web application development and web effort estimation. The results of the study, which calculated the size

of web application development using the WCP, Function Point (FP), and Web Object (WO) methods for ten projects, showed that the size calculated using the WCP method was in between the size calculated using the FP and WO methods. The future of this research holds significant promise, as it seeks to develop a web effort estimation model based on the proposed WCP method, thereby enabling a more accurate and efficient method of estimating the size of web applications. The ultimate goal is to empirically validate the effectiveness of the WCP method compared to the FP and WO methods, providing organizations with a valuable tool for better managing the complexity and cost of web application development. This research holds the potential to revolutionize the way web effort estimation is performed, offering greater accuracy and efficiency for organizations in the rapidly growing field of web development.

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Author Contributions Dr. Syed Mohsin Saif: Conceptualized the study, designed the research framework, and provided critical revisions to the manuscript. Dr. Syed Ansarullah: Assisted in the methodology development, conducted data analysis, and contributed to the writing and editing of the manuscript. Dr. Sami Alshmrany: Provided expert advice on the research topic and contributed to the interpretation of results. Anurag Sinha: [Corresponding Author] Coordinated the study, oversaw the project management, and ensured that all authors contributed to the final manuscript.

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Data Availability The data supporting the findings of this study are available from the corresponding author upon reasonable request. The datasets used during the study are not publicly available due to privacy concerns but can be accessed by researchers under specified conditions.

Declarations

Research Involving Human and/or Animals This research did not involve any studies on human participants or animals. All experiments and methodologies were conducted in accordance with ethical standards and relevant institutional guidelines.

Informed Consent As the study did not involve human participants, informed consent was not applicable. However, ethical approval was obtained from the relevant institutional review board for any potential future studies involving human subjects or animal research.

Competing Interests The authors declare that there are no competing interests that could potentially influence the findings or interpretations presented in this study. All authors have disclosed any financial or personal relationships that could be perceived as conflicts of interest related to the research.

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